

Ionic Bonding and Formula Units

Independent Application Answer Sheet - Notebook Evidence

Student name: _____ Date: _____

Directions

Open PhET Build an Atom and select the Atom screen. The simulation supplies no more than 10 protons, so build only the three ions shown below. Turn on Element, Net Charge, and Mass Number. Drag particles from the buckets, verify every displayed value, and record the evidence. Then use the charge-balance rule to write each formula unit. Submit this completed PDF in U3 L2 Notebook Evidence Submission.

Part A - Build and record ions

Model	Protons	Electrons	Net charge	Evidence
Lithium ion, Li ⁺	3	2		Element: Li; charge: +1
Fluoride ion, F ⁻	9	10		Element: F; charge: -1
Oxide ion, O ²⁻	8	10		Element: O; charge: -2

Part B - Balance charges to write formula units

Ions	Least whole-number ratio	Charge check	Formula unit
Na ⁺ and Cl ⁻			
Mg ²⁺ and Cl ⁻			
Al ³⁺ and O ²⁻			

Explanation

Explain why an ionic formula is a simplest whole-number ratio rather than a separate molecule. Use the terms ions, total charge, formula unit, and lattice.

Correction rule: If either total charge is not zero, change the ratio and recalculate before submitting.